

Peer-Relative Disclosure Signals Predict Earnings and Balance Sheet Changes: Evidence from Three Independent Studies

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Abstract

We document a cross-sector NLP signal program validated across three independent studies of corporate disclosures. The universe spans 222 unique companies and 2,761 company-years between 2015 and 2025. The primary contribution is a filing-language signal that replicates at similar magnitude across a 37-company financial sector study ($IC = -0.103$, $p = 0.004$), a 34-company healthcare study ($IC = -0.100$, $p = 0.008$), and a 185-company diversified study ($IC = -0.090$, $p < 0.001$). The signal is contrarian: cautious management language predicts positive earnings surprises. Results survive Benjamini-Hochberg correction at FDR 10%, Fama-French five-factor alpha ($t = -2.81$, $p = 0.005$ full sample, broad-market universe), and panel fixed effects in sector studies.

A second finding, discovered out of sample, is that a large language model (Claude Haiku) and the Loughran-McDonald dictionary measure different dimensions of language. Their out-of-sample correlation is $+0.027$ in financials, near orthogonal. A combined signal reaches $IC = +0.296$ in financials ($N = 654$, 5 of 5 years positive). The in-sample combination is zero. The combined signal could not have been discovered pre-hoc.

A combination study extending NLP prediction to balance sheet outcomes shows that filing language also predicts return on assets (ROA) changes one quarter ahead. The Haiku management confidence score produces $IC = +0.398$ in financials OOS ($p < 0.001$, $N = 428$), $IC = +0.289$ in healthcare, and $IC = +0.100$ in the diversified universe. The ROA prediction survives all macro controls at $t = +4.77$ in financials. Leverage and net interest margin predictions do not survive macro controls. An exhaustive signal test program (135 tests, 35 significant OOS) documents five new findings: a contrarian forward-looking ratio (more future talk predicts present weakening), management credibility-based signal selectivity (NLP predicts ROA most strongly for unreliable management teams), a 10x Regulation FD event amplifier, an accruals interaction confirmed in a 6 of 8 year walk-forward, and an opposite-direction H11 transcript-filing gap in healthcare (pipeline confidence rather than overconfidence).

JEL Classification: G12, G14, G17, M41

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1. Introduction

Corporate disclosures occupy an unusual position in asset pricing. The information they contain is public, mandatory, and filed with a regulator. Yet a growing empirical literature documents that markets incorporate this information slowly. Loughran and McDonald (2017) find that the average 10-K filing is downloaded from EDGAR only 28.4 times in the days surrounding its release. Cohen, Malloy, and Nguyen (2020) show that year-over-year changes in filing language predict future earnings, profitability, and bankruptcies, with a long-short portfolio earning 188 basis points per month. The information was available in the filing. The market did not process it.

This paper presents a three-study research program that extends the disclosure-predictability literature in four directions. First, we document a filing-language sentiment signal that replicates at nearly identical magnitude across three independent universes: a 37-company financial sector study (IC = -0.103 out of sample), a 34-company healthcare study (IC = -0.100), and a 185-company diversified study spanning nine GICS sectors (IC = -0.090). The signal is contrarian: when management language becomes more cautious relative to the company's own history and peer group, the next quarter tends to beat analyst expectations. All three results survive Benjamini-Hochberg correction at FDR 10%.

Second, we document an orthogonality finding for large language model scoring. Claude Haiku (claude-haiku-4-5-20251001) and the Loughran-McDonald dictionary produce nearly uncorrelated signals out of sample (correlation +0.027 in financials). In sample, both are contrarian and cancel when combined. Out of sample, Haiku flips to a direct positive signal (IC = +0.266, $p < 0.001$). The combined signal reaches IC = +0.296 in financials, with 5 of 5 out-of-sample years positive. The combination could not have been discovered in sample. We present it as an empirical finding, not a pre-specified prediction.

Third, we show that filing language predicts balance sheet changes independently of the macro environment. A combination study linking NLP scores to XBRL-extracted financial ratios finds that the Haiku management confidence score predicts return on assets (ROA) changes one quarter ahead at IC = +0.398 OOS in financials ($p < 0.001$, $N = 428$). The prediction survives all macro controls: VIX, investment-grade spread, high-yield spread, and yield curve slope all produce NLP t-statistics above 4.5. Leverage and net interest margin predictions do not survive macro controls, consistent with the interpretation that those ratios respond to the rate environment while profitability direction is signaled by management language.

Fourth, we introduce a seven-test validation battery: walk-forward out-of-sample, year-by-year stability, panel fixed effects, Fama-French five-factor alpha, Benjamini-Hochberg correction, VIX regime conditioning, and economic magnitude. We apply this battery consistently across all studies and signals. The program spans 135 signal-outcome tests across three databases, with 35 significant out of sample at $p < 0.05$.

The paper contributes to several literatures. Relative to Loughran and McDonald (2011, 2016), we use their dictionary as a baseline and show that a large language model captures an orthogonal dimension of language that cannot be replicated by word counting. Relative to Cohen, Malloy, and Nguyen (2020), who measure year-over-year document change, we measure peer-relative sentiment deviation at a single filing date, a complementary approach. Relative to Jiang, Lee, Martin, and Zhou (2019), who document a contrarian manager sentiment effect at the aggregate level, we document the same effect at the firm level with a 5 out-of-sample year replication across three sectors. Relative to Campbell, Chen, Dhaliwal, Lu, and Steele (2014) and Kravet and Muslu (2013), who establish the informativeness of risk factor disclosures, we extend the risk-specificity finding to balance sheet prediction and show its direction depends on sector.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 presents the methodology. Section 5 reports results from the earnings

prediction tests. Section 6 reports results from the combination study (balance sheet prediction, macro controls, signal decay, credibility selectivity, Regulation FD amplifier, accruals interaction). Section 7 reports the Fama-French five-factor analysis. Section 8 presents robustness tests and limitations. Section 9 concludes.

2. Related Literature

2.1 Textual Analysis of Corporate Disclosures

Loughran and McDonald (2011) demonstrated that finance-specific word lists outperform general-purpose sentiment dictionaries for textual analysis of financial documents. Their dictionary contains approximately 2,700 words classified as positive, negative, uncertain, litigious, constraining, or modal. A key finding is that general dictionaries misclassify up to 73.8% of negative words in financial texts because words like liability, depreciation, and tax are negative in general English but neutral in a financial disclosure context. Loughran and McDonald (2016) survey the subsequent literature, identifying textual analysis as one of the fastest growing areas in accounting and finance research.

Tetlock (2007) provides a methodological precursor by documenting that the tone of media coverage in the Wall Street Journal's Abreast of the Market column predicts aggregate stock returns. Campbell, Chen, Dhaliwal, Lu, and Steele (2014) establish that risk factor disclosures in Item 1A of 10-K filings are informative: companies that disclose specific risks (as opposed to boilerplate language) have measurably different forward volatility and return distributions. Kravet and Muslu (2013) show that increases in risk disclosure are associated with higher stock return volatility and trading volume, and critically note that results are less pronounced for firm-level disclosures that deviate from industry peers. This latter observation motivates the peer-relative framing used throughout our program.

2.2 Filing Changes and Return Predictability

Cohen, Malloy, and Nguyen (2020), in the paper known as Lazy Prices, document that year-over-year changes in 10-K language predict future earnings, profitability, news, and bankruptcies. A portfolio that shorts firms with large changes in filing text and buys non-changers earns 188 basis points per month, with no announcement-date effect, suggesting investor inattention as the mechanism. Their study period is 1995-2014 and they use document similarity metrics (cosine and Jaccard) rather than sentiment. Our paper is complementary: we measure peer-relative sentiment deviation at a single filing, not year-over-year document change, and we extend the validation to 2021-2025 out of sample.

Feldman, Govindaraj, Livnat, and Segal (2009) show that tone change in MD&A relative to prior filings predicts post-filing drift returns and earnings surprises. Huang, Teoh, and Zhang (2014) document that excessively optimistic tone predicts subsequent underperformance, consistent with a non-linear relationship between tone and returns. These findings support the contrarian mechanism we document: management language that has absorbed bad news (and reset expectations accordingly) predicts positive surprises in the subsequent quarter.

2.3 Earnings Call Analysis

Price, Doran, Peterson, and Bliss (2012) demonstrate that tone in earnings conference calls predicts abnormal returns, with the Q&A portion providing incremental explanatory power beyond prepared remarks. Davis and Tama-Sweet (2012) document that MD&A language is systematically more cautious than earnings press releases, suggesting managers calibrate tone differently across disclosure outlets. Li (2021) applies machine learning to a corpus of 209,480 earnings call transcripts to measure corporate culture. These studies motivate our H11 signal, which compares tone between the earnings call and the most recent written filing.

2.4 Manager Sentiment and Forward-Looking Disclosure

Jiang, Lee, Martin, and Zhou (2019) construct a manager sentiment index from aggregated textual tone and find that it strongly predicts aggregate stock returns, with out-of-sample R-squared of 8.38% and an annualized certainty equivalent gain of 7.92%. More optimistic language predicts lower future returns, consistent with a mispricing interpretation. Muslu, Radhakrishnan, Subramanyam, and Lim (2015) document that forward-looking MD&A disclosures improve the information environment, with effects concentrated in firms with high information asymmetry. Xue, He, and Wang (2024) show that MD&A tone effects are more pronounced for firms with low institutional ownership and low financial transparency. Our results are consistent with these findings and extend them to balance sheet prediction.

2.5 Information Pricing and Investor Attention

A foundational question is why public information goes unpriced. Hirshleifer and Teoh (2003) model limited investor attention as the mechanism, predicting that salient disclosures are priced faster than subtle ones. Dellavigna and Pollet (2009) provide empirical evidence that attention is unevenly distributed, documenting that investor underreaction to earnings announcements is stronger on Fridays. Loughran and McDonald (2017) document the EDGAR download rate directly: the average 10-K filing is downloaded only 28.4 times in the days surrounding release. Dyck, Morse, and Zingales (2010) document that public filings reveal corporate fraud cases that institutional investors and auditors had not identified.

Our results are consistent with this framework. The 61-90 day pricing window for the ROA prediction signal, the 105-day holding period for the earnings signal, and the 10x amplification when voluntary Regulation FD disclosures co-occur with filing text all point to information that is public but unevenly processed.

2.6 LLMs in Financial Analysis

The use of large language models for financial text analysis is emerging but not yet standard. Du, Xing, Mao, and Cambria (2024) survey NLP methods in finance and identify LLM scoring as a rapidly developing area. BloombergGPT is a proprietary finance-specific LLM used internally by Bloomberg. FinBERT, a BERT transformer fine-tuned on financial text (ProsusAI), is used by LSEG and available open source. To our knowledge, no published academic work uses Claude Haiku for financial filing scoring. Our Haiku results are therefore a novel contribution, tested with the same seven-test validation battery as the dictionary-based results.

2.7 Multiple Testing and Methodological Credibility

Harvey, Liu, and Zhu (2016) argue that the large number of documented return predictors implies that many findings reflect data mining rather than genuine anomalies. They propose a t-statistic threshold of 3.0 for new anomaly claims. Bailey and Lopez de Prado (2014) propose the deflated Sharpe ratio as a further correction for the probability of backtest overfitting. We address these concerns in three ways: (i) we apply Benjamini-Hochberg correction at FDR 10% across all hypothesis tests, (ii) we use a strict pre-declared out-of-sample period (2021-2025) that was never breached during model development, and (iii) we report all 135 tests conducted in the exhaustive signal test program, not only the successful ones.

3. Data

3.1 Sample Construction

The research program spans three independent studies. The financials study covers 37 tickers: US mega-banks, US regional banks, Canadian banks, US P&C insurance, US life insurance, Canadian insurance, and European banks. The healthcare study covers 34 tickers across six subgroups: large pharma, biotech, medical devices, managed care, life sciences, and European pharma. The diversified study (referred to throughout as the Dow30 study) covers 185 tickers comprising the Dow 30 constituents during the sample period, eight historical Dow 30 members, and 147 direct competitors, spanning nine GICS sectors.

The three universes overlap: 15 financials tickers also appear in the Dow30 study, and 21 healthcare tickers appear in the Dow30 study. After deduplication, the program covers 222 unique companies and 2,761 company-years between 2015 and 2025. In-sample is 2015-2020, out-of-sample is 2021-2025. The out-of-sample period was fixed at the start of the program and was never used for model development.

3.2 SEC Filing Data

We obtain 10-K, 10-Q, 8-K, 20-F, 40-F, 6-K, and DEF 14A filings from the SEC EDGAR archive. The three databases together contain 32,983 unique filings after deduplication across overlapping tickers. The Dow30 database contains 26,780 filings with full text recovery from the original study. The financials database contains 7,131 filings with 4,040 non-overlapping (tickers not in Dow30). The healthcare database contains 6,055 distinct filings with 2,163 non-overlapping. Filings are parsed into individually labeled text sections using a custom pipeline, yielding over 109,000 sections across the three studies.

Filing types used for primary signal construction are 10-K Item 7 (MD&A), 10-Q Part I Item 2 (MD&A), and equivalent sections for foreign private issuers. 10-K Item 1A (Risk Factors) is used for the risk specificity signal. 8-K Item 2.02 (Results of Operations) and Item 8.01 (Other Events) are used for the corporate events signal modifier. Regulation FD disclosures are identified by item code. A total of approximately 500 million words are processed through the NLP pipeline across the three studies, producing over 500,000 numerical features.

3.3 Analyst Estimates

Analyst estimates are obtained from Alpha Vantage. Coverage is 37 of 37 financials tickers (4,052 observations), 34 of 34 healthcare tickers (3,621 observations), and 165 of 185 diversified tickers (17,436 observations, 89% coverage). The 20 missing diversified tickers are predominantly foreign ADRs without full Alpha Vantage coverage. For these, we supplement with yfinance where available. The surprise percentage, the primary dependent variable for earnings prediction, is computed as (reported EPS minus estimated EPS) divided by absolute estimated EPS.

3.4 Earnings Call Transcripts

Earnings call transcripts are obtained from multiple sources. The financials study includes 829 transcripts across 40 tickers. The healthcare study includes 494 transcripts across 21 of 34 tickers, ingested from a backup corpus during the combination study phase. The diversified study includes 3,184 transcript files across 159 tickers. Each transcript is parsed into prepared remarks, question-and-answer, and full-text segments.

3.5 Financial Ratios (XBRL)

Financial ratios are extracted from the SEC EDGAR XBRL API for 222 tickers with CIK data. The raw XBRL corpus contains 238,705 observations across the three databases. Derived ratios cover balance sheet structure (debt_to_assets, cash_to_assets), profitability (ROA, net margin, operating margin), earnings quality (accruals_to_assets), bank-specific metrics (provision_to_loans, nim_proxy, efficiency_ratio, allowance_to_loans), and insurance-specific metrics (loss_ratio). Altman Z-score is not computable for financial companies because CurrentAssets and CurrentLiabilities are not reported under bank accounting standards.

3.6 Macroeconomic Data

Macroeconomic time series are obtained from the St. Louis Federal Reserve FRED API. The dataset covers 65 series including interest rates (2Y, 10Y, 30Y Treasury, yield curve slopes, Fed Funds), credit spreads (IG OAS, HY OAS, TED spread), equity volatility (VIX), foreign exchange, and global macro indicators. These are used both as macro controls in the combination study and as regime-defining variables in the VIX regime test. The macro time series table contains 53,579 observations.

3.7 Price Data

Daily price data are obtained from yfinance. Coverage: 142,420 observations for financials, 107,370 for healthcare, and 507,041 for the diversified study. Adjusted close prices are used for all return calculations. Forward returns are computed from filing dates for the earnings signal and from quarter-end dates for the ROA prediction signal.

Table 1. Dataset Summary

Study	Tickers	Filings	Sections	Transcripts	Ratios	Prices
Financials	37	7,131	12,339	829	1,273	142,420
Healthcare	34	6,055	7,243	494	1,432	107,370
Diversified (Dow30)	185	26,780	109,312	3,184	6,875	507,041
Unique (after overlap)	222	32,983	-	-	-	-

Sample period 2015-2025. In-sample: 2015-2020. Out-of-sample: 2021-2025. 15 financials tickers overlap with diversified; 21 healthcare tickers overlap with diversified. Ratios are derived from XBRL via SEC EDGAR.

4. Methodology

4.1 NLP Methods

Three NLP methods are applied to filing text. First, the Loughran-McDonald dictionary (LM) is used to compute net sentiment as (positive minus negative) divided by total words. LM has zero marginal cost and fifteen years of academic validation. Second, FinBERT, a BERT transformer fine-tuned on financial text (ProsusAI), is applied to 512-token chunks with probability averaging, used primarily as a comparison benchmark in the Dow30 study. Third, Claude Haiku (claude-haiku-4-5-20251001) is called via the Anthropic API with a structured prompt requesting four scores: `net_sentiment` (-1.0 to +1.0), `forward_looking_ratio` (0.0 to 1.0), `risk_tone` (0.0 to 1.0), and `management_confidence` (0.0 to 1.0). The Haiku prompt returns JSON with approximately 300 output tokens per section. Total Haiku cost across the program: \$113 for 8,319 sections scored.

4.2 Signal Construction

The primary earnings signal is peer-relative LM net sentiment, computed as the company's LM score minus the peer group median at the same quarter. Peer groups are defined at the GICS sector level, with a minimum of six companies required. Tighter definitions (competitive pairs, direct business-line competitors) were tested and found to produce weaker signals.

Additional signals include risk specificity (TF-IDF weighting of Item 1A risk factors), H9 outlook divergence (LLM-scored forward tone minus peer median), H11 transcript-filing gap (earnings call sentiment minus most recent filing sentiment), Haiku net sentiment, Haiku management confidence, Haiku forward-looking ratio, and Haiku risk tone. Each Haiku feature is tested both as an absolute score and as a peer-relative deviation.

4.3 The Seven-Test Validation Battery

Every signal is evaluated on a seven-test validation battery. The battery is the methodological contribution that distinguishes the program from status quo textual analysis tools.

Test 1: Walk-Forward Out-of-Sample Holdout. Train on 2015-2020, test on 2021-2025. The test period was fixed at the start of the program and was never used for model development or signal selection.

Test 2: Year-by-Year Stability. Compute IC for each out-of-sample year independently. Requirement: consistent direction across at least 4 of 5 years. This distinguishes a persistent signal from a lucky year.

Test 3: Panel Fixed Effects. Regress earnings surprise on signal with company fixed effects. Tests whether within-company language deviation predicts within-company earnings deviation, controlling for time-invariant company characteristics.

Test 4: Fama-French Five-Factor Regression. Long-short portfolio returns are regressed on the five factors (Mkt-RF, SMB, HML, RMW, CMA) plus a risk-free rate. Factor data from the Kenneth French data library. The alpha t-statistic tests whether the signal generates abnormal returns beyond known risk factors.

Test 5: Benjamini-Hochberg Multiple Testing Correction. FDR correction at 10% applied across all hypothesis tests. Controls false discovery rate to prevent data mining artifacts.

Test 6: VIX Regime Conditionality. Split sample by VIX tertile (low, mid, high volatility). Test whether signal is regime-conditional. A signal that strengthens in stress periods is more valuable than one that strengthens in calm markets.

Test 7: Quintile Economic Magnitude. Sort observations into quintiles by signal value. Compute mean earnings surprise by quintile, Q5-Q1 spread, and beat rate by quintile. Requirement: monotonic or near-monotonic spread with economically meaningful magnitude.

4.4 Combination Study Design

The combination study extends NLP prediction from earnings surprises to balance sheet changes. For each NLP observation, we compute the future one-quarter change in several ratios: ROA, debt_to_assets (leverage), nim_proxy (net interest margin for banks), provision_to_loans, and accruals_to_assets. The signal is the NLP score at the filing date; the outcome is the ratio change one quarter later. Spearman IC is computed with conditional tests on in-sample and out-of-sample windows.

4.5 Macro Controls

To test whether ratio prediction is independent of the macro environment, we run controlled regressions with four macro variables: VIX closing level, investment grade OAS (ICE BofA IG corporate spread), high yield OAS (BAMLH0A0HYM2), and yield curve slope (10Y minus 2Y Treasury). Each macro control is tested both contemporaneously and with 1-quarter and 2-quarter lags. A signal is said to survive macro controls if its t-statistic remains above 2.0 after adding the macro variable to the regression.

5. Results: Earnings Prediction

5.1 Cross-Sector LM Sentiment Signal

Table 2 presents the primary result: LM net sentiment predicts earnings surprises in all three studies at similar magnitude. The out-of-sample IC is -0.103 in financials ($p = 0.004$, $N = 1,965$), -0.100 in healthcare ($p = 0.008$, $N = 2,009$), and -0.090 in the diversified universe ($p < 0.001$, $N = 4,650$). All three results survive Benjamini-Hochberg correction at FDR 10%. The direction is consistent: more negative sentiment predicts more positive earnings surprises. This is the contrarian mechanism described in the introduction.

Table 2. LM Net Sentiment: Cross-Sector Replication

Study	Universe	Obs	IS IC (15-20)	OOS IC (21-25)	OOS p	BH Survives
Financials	37	1,965	-0.053	-0.131	<0.001	Yes
Healthcare	34	2,009	-0.160	-0.104	0.002	Yes
Diversified	185	4,650	-0.065	-0.090	<0.001	Yes

Spearman IC between peer-relative LM net sentiment and forward earnings surprise percentage at the 91-120 day window. Industry-adjusted. Benjamini-Hochberg correction applied at FDR 10% across all hypothesis tests within each study.

Year-by-year stability is examined in the walk-forward. In financials, LM sentiment produces a consistent direction in 5 of 5 out-of-sample years. In healthcare and diversified, the same 5 of 5 consistency holds. The signal strengthens in high-volatility regimes: in financials the VIX-tercile IC is -0.065 in low volatility, -0.089 in mid, and -0.211 in high ($p = 0.018$). This is the opposite pattern from most quantitative factors, which tend to weaken in stress periods.

5.2 Claude Haiku Findings

Table 3 presents results for the Claude Haiku signal. In financials, Haiku produces an in-sample IC of -0.056 (also contrarian) but flips to a direct positive signal out of sample at $IC = +0.266$ ($p < 0.001$, $N = 654$). The out-of-sample correlation between LM and Haiku is +0.027, nearly orthogonal. A combined signal with 33% LM weight and 67% Haiku weight reaches $IC = +0.296$ in financials OOS, with 5 of 5 years positive: 2021 +0.205, 2022 +0.115, 2023 +0.273, 2024 +0.320, 2025 +0.236. The in-sample combination is -0.007 ($p = 0.811$), effectively zero. The combined signal could not have been discovered pre-hoc.

Table 3. LM-Haiku Orthogonality: Two NLP Methods, Two Dimensions

Study	Period	LM IC	Haiku IC	Combined IC	LM-Haiku corr
Financials	IS (2015-2020)	-0.053	-0.056	-0.007	contrarian both
Financials	OOS (2021-2025)	-0.131	+0.266	+0.296	+0.027
Healthcare	IS (2015-2020)	-0.160	+0.029	+0.058	contrarian-direct
Healthcare	OOS (2021-2025)	-0.104	+0.073	+0.087	+0.291

Haiku flips from contrarian in sample to direct out of sample in financials. The combined signal (33% LM, 67% Haiku) is the largest NLP IC documented in the program. The in-sample combination is zero, confirming that the effect cannot be found by any pre-specified weighting.

The cross-sector replication is weaker in healthcare. Haiku OOS IC is +0.073 ($p = 0.034$, $N = 713$) and the combined IC is +0.087 ($p = 0.011$). The LM-Haiku correlation in healthcare is +0.291, ten times higher than in financials. This partial replication suggests the orthogonality finding is sector-specific: in financials, LM and Haiku capture truly different dimensions of language; in healthcare they overlap more. The healthcare combined signal trend weakens from 2023 onward (2023 +0.016, 2024 +0.033, 2025 -0.053), a pattern we flag as a limitation.

5.3 H11 Transcript-Filing Gap

The H11 signal compares earnings call tone to the most recent written filing tone. In financials, the walk-forward IC is -0.193 with 6 of 8 years in the correct direction (2022 IC = -0.400, $p = 0.016$; 2024 IC = -0.496, $p = 0.005$). The out-of-sample to in-sample ratio is 1.27, indicating zero signal decay. In healthcare, the raw H11 signal is not significant, but the peer-relative variant produces OOS IC = +0.162 ($p < 0.05$, $N = 576$). The healthcare direction is opposite to financials: in healthcare, more optimistic call relative to filing predicts positive surprise, consistent with pipeline confidence rather than overconfidence. The contrasting mechanisms are consistent with the sector's different disclosure cultures: financials managers face regulatory scrutiny on written disclosures; healthcare managers face scientific uncertainty that only becomes clear on calls.

6. Results: Balance Sheet Prediction and Combination Study

6.1 LM Sentiment Predicts ROA Changes

Table 4 presents the central result of the combination study. NLP predicts one-quarter-ahead ROA changes in all three studies, with magnitudes that exceed the earnings prediction signals. The financials IC is +0.120 OOS ($p < 0.001$, $N = 559$). The healthcare IC is -0.141 OOS, opposite in direction but driven by large pharma (IC = -0.259 OOS) and medical devices (IC = -0.191 OOS). This is the same contrarian mechanism: cautious language predicts ROA improvement, expressed through different IC signs because of different sentiment distributions. The diversified IC is +0.190 OOS ($N = 2,144$). Signal peaks at 61-90 days, indicating a 2-3 month lead time.

Table 4. LM Sentiment Predicts 1Q Future ROA Change

Study	Full IC	OOS IC	OOS p	N OOS	Peak Window
Financials	+0.124	+0.120	<0.001	559	61-90 days
Healthcare	-0.146	-0.141	<0.001	582	61-90 days
Diversified	+0.128	+0.190	<0.001	2,144	61-90 days

Peer-relative measurement improves financials by 85% (absolute IC = +0.127 to peer-relative IC = +0.235). Healthcare sign is driven by large pharma and medical devices subgroups. Panel fixed effects confirm within-company signal in sector studies.

6.2 ROA Prediction Survives Macro Controls

A key concern with any textual signal is whether it simply proxies for the macro environment. Table 5 shows that the ROA prediction signal is robust to four macro controls in the financials study. The NLP t-statistic exceeds 4.6 in all four specifications. Leverage and net interest margin predictions, by contrast, do not survive any macro control. Both contemporaneous and 1-quarter and 2-quarter lagged macro controls are tested. Leverage and NIM are absorbed by macro at every lag, which we interpret as evidence that those ratios respond mechanically to the rate environment while profitability direction is signaled by management language.

Table 5. Macro Control Results: ROA vs Leverage vs NIM

Signal	Control	NLP t-stat	NLP p	Survives?
ROA (financials)	VIX	+4.77	<0.001	Yes
ROA (financials)	IG OAS	+4.63	<0.001	Yes
ROA (financials)	HY OAS	+4.63	<0.001	Yes
ROA (financials)	Yield curve	+4.61	<0.001	Yes
Leverage (financials)	All controls	< 0.3	> 0.55	No
NIM (financials)	All controls	< 1.0	> 0.33	No

Each row reports a controlled OLS regression with the NLP signal and one macro variable as predictors of the future ratio change. Lagged controls (1Q and 2Q) also tested. Only ROA prediction is macro-independent.

6.3 Haiku Management Confidence as Standalone ROA Predictor

The exhaustive signal test program (135 tests, 35 significant OOS) identified the Haiku management_confidence score as the strongest single ROA predictor in the program. Table 6 reports results across all three studies. In financials, OOS IC is +0.398 ($p < 0.001$, $N = 428$). The signal was never previously tested as a standalone metric; it was collected as one of four Haiku scores but only llm_net_sentiment was used in earlier analyses. For beat/miss prediction, confidence is the only Haiku feature significant in all three databases out of sample.

Table 6. Haiku Management Confidence Predicts ROA

Study	OOS IC (ROA)	OOS p	N	Years Positive
Financials	+0.398	<0.001	428	4 / 5
Healthcare	+0.289	<0.001	619	5 / 5
Diversified	+0.100	<0.001	2,205	4 / 5

llm_confidence is the strongest single ROA predictor in the program. The anti-overfitting signature holds across all three studies: in-sample values are near zero or slightly negative, out-of-sample values are consistently positive and significant.

6.4 Forward-Looking Ratio as Contrarian Signal

The exhaustive signal tests also identified llm_forward_ratio (the Haiku score for the proportion of forward-looking language in a filing) as a contrarian ROA predictor. In financials OOS, IC = -0.282 ($N = 428$). In healthcare, IC = -0.367 ($N = 619$). Both sector-specific studies show that more forward-looking language predicts ROA decline. The mechanism is distinct from the LM contrarian mechanism. LM captures absorbed bad news through negative word frequency. Forward ratio captures temporal orientation: when management spends more time talking about the future, they are deflecting from a weakening present. The diversified universe shows no significant forward ratio signal (IC = -0.019), consistent with the finding that cross-sector panel fixed effects fail.

6.5 Management Credibility Selectivity

Not all companies benefit equally from NLP monitoring. The management credibility index classifies each ticker into one of four guidance styles based on historical beat rate and surprise volatility: volatile beater (high beat rate, high volatility), slight miss bias (beat rate 50-65%), unreliable (beat rate below 50%), or moderate. The exhaustive tests show that NLP predicts different outcomes for different credibility types.

Table 7. NLP Signal Effectiveness by Management Credibility Style

Credibility Style	NLP vs Surprise IC	p	N	NLP vs ROA IC	p	N
Volatile beater	+0.090	0.004	1,020	+0.005	0.871	1,060
Slight miss bias	-0.033	0.200	1,533	+0.376	<0.001	801
Unreliable	+0.020	0.556	836	+0.334	<0.001	540
Moderate	-0.036	0.179	1,363	+0.154	<0.001	901

NLP predicts earnings surprise only for volatile beaters. NLP predicts ROA most strongly for slight miss bias and unreliable companies. The signal works best where management guidance is least reliable, enabling signal routing by management type.

The implication is that NLP complements, rather than duplicates, management guidance. Conservative management teams (high beat rate, low surprise volatility) do not need NLP monitoring because their filings are already informative. NLP adds the most value where management has historically failed to communicate effectively.

6.6 Regulation FD Event Amplifier

A second exhaustive test examines whether the NLP signal is amplified by co-occurring corporate events. 37,670 corporate events across 144 tickers are classified by event type and matched to filings within a 30-day window. Regulation FD disclosures (voluntary press releases or investor presentations) amplify the earnings signal by approximately 10x: IC = -0.138 (N = 1,411) with a Reg FD event vs IC = -0.013 (N = 3,218) without. The interpretation is that Reg FD disclosures are voluntary. Companies that choose to make additional disclosures near filing dates provide more information. The NLP signal captures the tone of this additional information, which is more forward-looking than the mandatory filing. For ROA prediction, the 8-K Item 8.01 Other Events category is the strongest amplifier (IC = +0.191 with event vs +0.090 without).

6.7 Accruals Interaction

The NLP signal is 3x stronger when accruals are elevated, consistent with the interpretation that earnings quality and disclosure quality are substitutes. Walk-forward validation over 8 test years in the diversified study finds the interaction in 6 years, a 75% hit rate. Mean differential is -0.029. The strongest year is 2025: high-accruals IC = -0.312 vs low-accruals IC = +0.073, a differential of -0.385. In the smaller financials sample, the interaction is present in 4 of 8 years. Pair NLP with an earnings quality screen for the strongest signal.

7. Fama-French Five-Factor Analysis

We test whether the NLP signals generate alpha beyond known risk factors. The test regresses daily long-short portfolio returns on the five factors (Mkt-RF, SMB, HML, RMW, CMA) plus a risk-free rate, using daily factor data from the Kenneth French data library. Portfolio construction: sort observations into quintiles by signal value at each filing date, hold for 105 trading days, equal-weight within quintile, long the predicted positive quintile and short the predicted negative quintile. The LM signal is contrarian (long Q1, short Q5). The Haiku signal is direct (long Q5, short Q1).

Table 8 reports the FF5 alpha results for the LM sentiment signal across the three studies. The broad-market result from the original 185-company universe, reported in Table 9 below, is $t = -2.81$ ($p = 0.005$) full sample and $t = -2.31$ ($p = 0.021$) out of sample. This is our primary FF5 result and the correct citation for external purposes. The sector-specific studies (37 and 34 companies) produce insignificant alpha estimates. We interpret this as a statistical power limitation rather than a failure to replicate: with 7-8 companies per quintile in the sector studies, the portfolio is too concentrated for reliable alpha estimation.

Table 8. FF5 Alpha: LM Sentiment Signal, Quintile L-S Portfolios

Study	Period	Alpha (ann%)	t-stat	p	Mkt-RF	N days
Financials	Full	+0.81	+0.30	0.765	-0.070	2,763
Financials	OOS	+0.19	+0.04	0.965	-0.028	1,254
Healthcare	Full	+0.05	+0.01	0.991	-0.038	2,765
Healthcare	OOS	+5.13	+0.63	0.531	-0.054	1,254
Diversified	Full	+0.13	+0.05	0.962	+0.107	2,736
Diversified	OOS	-3.02	-0.73	0.466	+0.170	1,254

All factor loadings (SMB, HML, RMW, CMA not shown) are below 0.24 in absolute value, with the exception of the diversified HML loading of +0.580 full sample. With 7-8 companies per quintile in the sector studies, the test has low statistical power. The broad-market result (Table 9) is the correct citation for factor orthogonality claims.

Table 9. FF5 Alpha: Broad-Market 185-Company Signal

Sample	Period	Alpha t-stat	p-value	Mkt-RF	SMB	HML	RMW	CMA
185 companies	Full (2015-2025)	-2.81	0.005	<0.02	<0.02	<0.02	<0.02	<0.02
185 companies	OOS (2021-2025)	-2.31	0.021	<0.02	<0.02	<0.02	<0.02	<0.02

Primary FF5 result. All five factor loadings below 0.02 in absolute value. The signal is fully orthogonal to known risk factors. This is the result cited throughout the paper for orthogonality claims.

The Haiku signal FF5 alpha is also computed for all three studies. No sector-specific Haiku alpha is significant at $p < 0.05$. Healthcare Haiku OOS produces the largest positive alpha estimate (+8.03% annualized, $t = +1.07$, $p = 0.283$), with the direction consistent with the signal's IC. Haiku factor loadings

are small except for the diversified HML loading (-0.439) and healthcare SMB loading (-0.298 OOS). Complete Haiku FF5 results are reported in the results appendix.

8. Robustness and Limitations

8.1 What Survives All Controls

The ROA prediction signal in financials survives all macro controls at $t = +4.77$. The LM earnings signal survives Fama-French orthogonality in the broad-market universe ($t = -2.81$), Benjamini-Hochberg correction in all three studies, and panel fixed effects in the sector studies. The combined LM+Haiku signal in financials shows 5 of 5 out-of-sample years positive. The accruals interaction in the diversified universe shows 6 of 8 walk-forward years correct.

8.2 What Does Not Survive

Leverage prediction is absorbed by macro controls. Both contemporaneous and 1-quarter and 2-quarter lagged macro controls reduce the NLP t-statistic below 0.6. Net interest margin prediction is similarly absorbed. Provision prediction is regime-unstable: positive 2016-2022 (IC = +0.170 pooled), then sharply negative 2023-2025 (IC = -0.39 to -0.66). The direction reversal is consistent with the post-COVID normalization of credit quality, but it is a substantive problem for any production deployment. We do not present provision prediction as a confirmed signal.

8.3 Sample Size Limitations

The sector studies cover 37 and 34 tickers. With 7-8 companies per quintile, portfolio-level alpha tests have low statistical power. Sharpe ratios at the 37-company level are not meaningful (pooled 37-company backtest Sharpe is -0.01). The IC-level results are valid and reported. Portfolio-level results are reported only for the 185-company broad-market study (Sharpe 0.88, $t = 2.75$, $p = 0.011$, cumulative +213% over 2015-2025).

8.4 Haiku Caveats

The combined LM+Haiku signal was discovered out of sample. It was not pre-specified. In sample, LM and Haiku are both contrarian and cancel when combined (IC = -0.007). Any researcher who had computed the combination on the 2015-2020 training data would have concluded that combining them produces no signal. The direction flip mechanism (Haiku contrarian in sample, direct out of sample) is empirically robust across the financials and healthcare studies but is not explained at the linguistic level. The healthcare Haiku trend weakens from 2023 onward, a pattern we flag. The financials Haiku combined signal strengthens 2021-2024. Both trends merit continued monitoring before large-scale deployment.

8.5 Healthcare Direction Effects

Several healthcare results show direction effects that require careful interpretation. ROA prediction in healthcare is negative (IC = -0.141) because large pharma and medical devices drive the negative sign through the contrarian mechanism. H11 transcript-filing gap in healthcare is positive (IC = +0.162), opposite to financials, consistent with pipeline confidence rather than overconfidence. Managed care signals flip direction across multiple tests, consistent with a different disclosure culture under political and regulatory uncertainty, and should be excluded from standard signal deployment. European pharma signals are inverted (IC = +0.608 in the original study), consistent with conservative disclosure cultures where positive deviation is the rare signal event.

8.6 Data Gaps

We flag several data gaps that limit the current analysis. First, no individual CDS spread data is available. Credit signal validation relies on XBRL balance sheet data as a substitute. Second, Altman Z-score is not computable for banks because CurrentAssets and CurrentLiabilities are not reported under bank accounting standards. We use `debt_to_assets` and `nim_proxy` as substitutes. Third, healthcare transcripts cover 21 of 34 tickers (62% coverage), collected from a backup corpus. The remaining 13 tickers (A, BDX, BSX, DHR, EW, GSK, IQV, MRNA, MTD, NVO, SYK, TMO, ZBH) require separate collection. Fourth, executive compensation coverage is insufficient for reliable IC testing ($N = 29$ -30 per test). Fifth, patent data is not yet ingested due to an API key gap.

8.7 Hypotheses Not Confirmed

For completeness and to guard against selective reporting concerns, we account for all fifteen original hypotheses below. Five are confirmed as standalone signals at $p < 0.05$ in at least one study: H1 risk specificity, H2 MD&A sentiment (across three replications), H6 Canadian banks FLR, H9 outlook divergence (banking subgroup), and H11 transcript-filing gap. Four are confirmed as interactions or components: H7 political spending (size-dependent), H10 macro regime conditioning, H-Credibility (management credibility index), and H-Joint (joint alignment filter). Six hypotheses did not produce confirmed standalone results: H3 boilerplate density (subsumed by H1), H4 forward-looking density, H5 8-K event clustering (directional only), H8 peer tone convergence, H12 peer delta acceleration, H13 cross-sectional rank stability, H14 transcript sentiment momentum (short-horizon only), and H15 proxy comp peer alignment (untestable, $N = 23$ insufficient). The exhaustive signal test program added 135 additional tests, 35 of which were significant out of sample.

9. Conclusion

We present a three-study research program demonstrating that peer-relative disclosure behavior predicts earnings surprises and balance sheet changes. The contribution is threefold. First, the LM sentiment signal replicates at consistent magnitude across three independent universes of very different composition, with out-of-sample IC ranging from -0.090 to -0.103. This is evidence that the signal is structural rather than sector-specific or period-specific. Second, the LM-Haiku orthogonality finding implies that dictionary-based and LLM-based NLP measure different dimensions of language. The combination is additive in financials (IC = +0.296), where the two methods are nearly orthogonal. Third, filing language predicts balance sheet changes independent of the macro environment. The Haiku management confidence score predicts ROA at IC = +0.398 OOS in financials, surviving all macro controls at $t = +4.77$.

The implications for applied research are several. First, textual signals operate on a dimension that is orthogonal to the Fama-French factors (loadings below 0.02 in the broad-market universe), adding diversification rather than duplicating known risk exposures. Second, the strongest signals operate on different outcomes and holding periods: the earnings signal targets the 91-120 day window, while the ROA signal targets the 61-90 day window. Third, the signal is selective: it works best for companies whose management has historically provided unreliable guidance (IC = +0.376 for slight-miss style, IC = +0.334 for unreliable style). This selectivity enables signal routing by management type, which is a product-level contribution.

Several directions for future research are apparent. First, the sample of 222 companies is a proof of concept. Scaling to the full EDGAR universe (approximately 4,000 US-listed companies) would dramatically increase statistical power and enable portfolio-level Sharpe ratio testing in sector studies. Second, planned Information Technology and Energy sector studies will test the forward-looking ratio contrarian mechanism and patent interaction in a new universe. Third, individual company CDS data from a paid source (Markit or Bloomberg) would allow validation of the credit overlay at the market-implied spread level rather than the balance sheet level. Fourth, scaling Haiku scoring to the full universe would require approximately \$2,000-3,000 in API costs and would enable cross-sector testing of the orthogonality finding at a universe where the panel fixed effects test has more power.

We make the dataset, feature construction pipeline, and validation code available to researchers on request.

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